

EXPERIMENT 8

Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

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Title

Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application – Vehicle Service Request (VSR) Application.

Aim

To configure user permissions, roles, role hierarchy and data sharing rules in the Vehicle Service Request application developed in Zoho Creator, and to study how role-based access control regulates access to application data.

Objective

To configure user management, user permissions, roles, role hierarchy and data-sharing rules in a Zoho Creator application and understand how role-based access control can control access to application data.

Software / Tools Required

- Zoho Creator (web-based low-code application development platform)
- Vehicle Service Request (VSR) application created in Zoho Creator
- Web browser with an active internet connection
- Zoho account with access to the application in Edit mode

Introduction / Theory

User Management in Zoho Creator

User management in Zoho Creator refers to the set of features used to decide who can access an application and what each person is permitted to see and do within it. Users are added to an application and are associated with permission sets (profiles) and roles. Together, these settings control access to the forms, reports, pages and records of the application. In the Settings area of an application, the User Permissions section groups these controls under three tabs: Permissions, Roles and Data Sharing.

Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is an access-control model in which access rights are assigned to roles instead of to individual users, and users acquire rights through the roles they hold. RBAC simplifies administration, supports the principle of least privilege (users receive only the access necessary for their duties), improves data security and keeps the access policy consistent and easy to audit.

Key Concepts

Concept	Description
Users	Individuals who are added to the application and who sign in to work with its forms, reports and data.
Permissions	Named permission sets (profiles) that define what a user may do with application components, for example Read or Write.
Roles	Positions in the organisational structure of the application (for example CEO). A role determines the position of a user in the hierarchy and the visibility of records.
Role Hierarchy	A tree structure in which each role reports to another role. A role higher in the hierarchy can generally view the records of the roles that report to it, whereas the reverse does not apply.
Data Sharing	Share rules that extend record visibility beyond the default hierarchy by sharing the records of a form from one role or user to another role or user with a defined level of access (for example Read Only or Read/Write).

Procedure

1. Sign in to Zoho Creator. The home page lists the applications available in the account. Locate the Vehicle Service Request application.

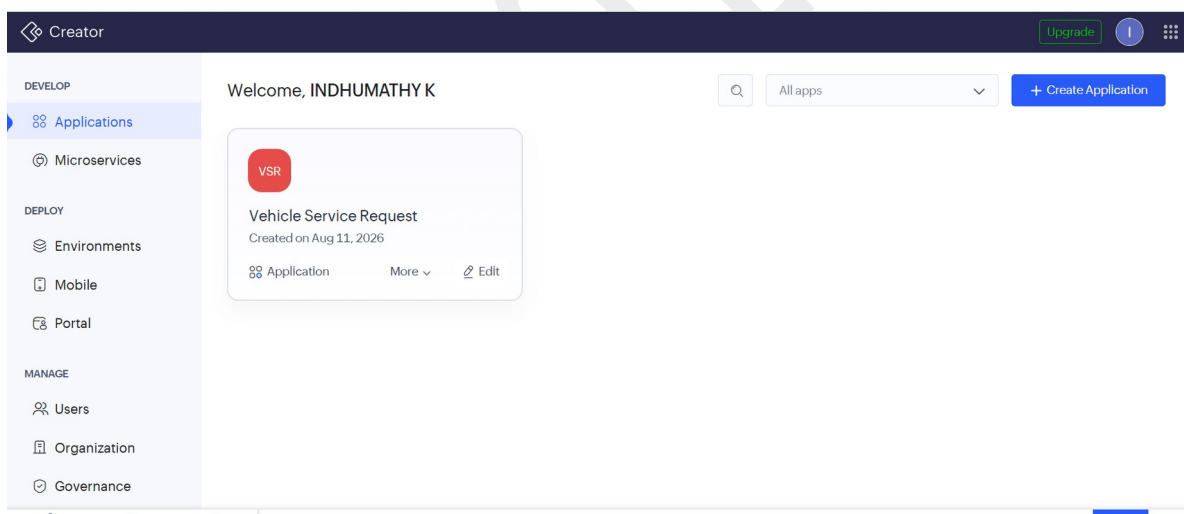


Figure 8.1: Zoho Creator Home – Vehicle Service Request Application

2. Open the application to view its dashboard (Home page), which contains the Request Kanban, All Requests, New Request and Vehicle Service Requests menu items, summary tiles, charts and the All Request Report.

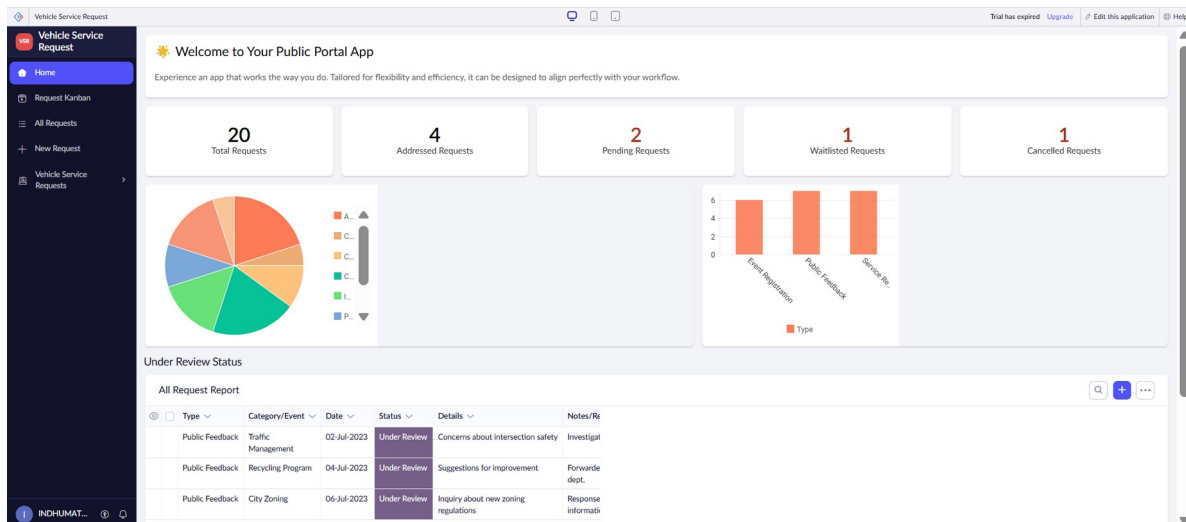


Figure 8.2: Vehicle Service Request Application – Access Mode

- Enter Edit mode by selecting Edit on the application card (Figure 8.1) or Edit this application at the top-right corner of the application (Figure 8.2).
- Navigate to Settings:** In Edit mode, select the Settings tab from the top navigation bar (Design, Workflow, Settings).
- Open User Permissions:** Under the PERMISSIONS group of the left-hand panel, select User Permissions. The Users page, which is used to add and manage the permissions and roles of users and to define record visibility, is displayed with the tabs Permissions, Roles and Data Sharing.
- Study the Permissions tab:** The Permissions tab lists the available permission sets. Two permission sets are available – Read (read permission for all components) and Write (write permission for all components). An Add Permission button is also provided.

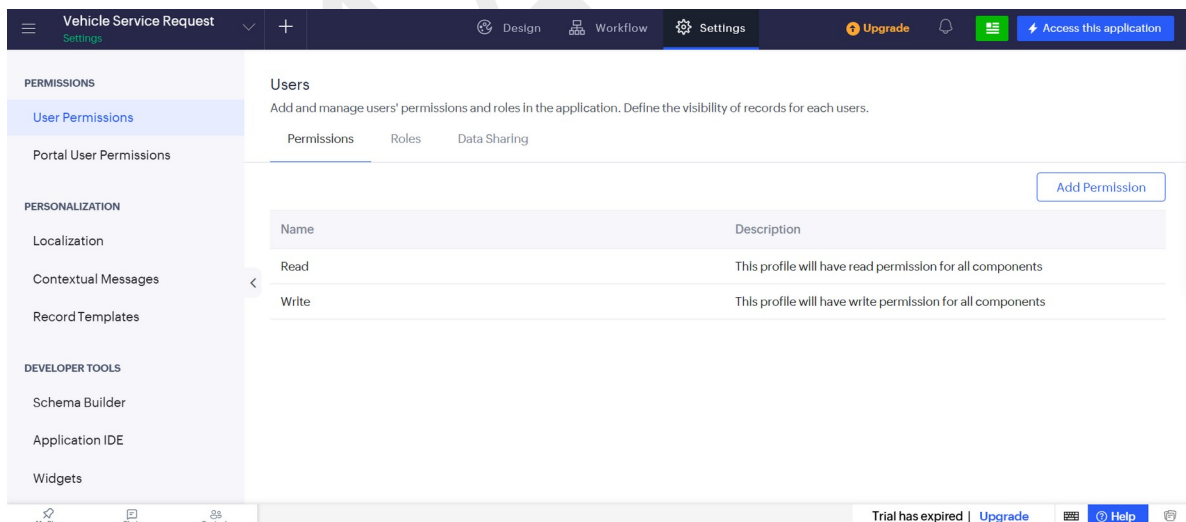


Figure 8.3: User Permissions – Permissions Section

- Study the Roles tab:** Select the Roles tab. The default role, CEO, is displayed at the top of the role tree. The Expand All, Collapse All, Manage Role Hierarchy and Add Role options are available. The Role Hierarchy interface is used to arrange roles in a reporting structure.

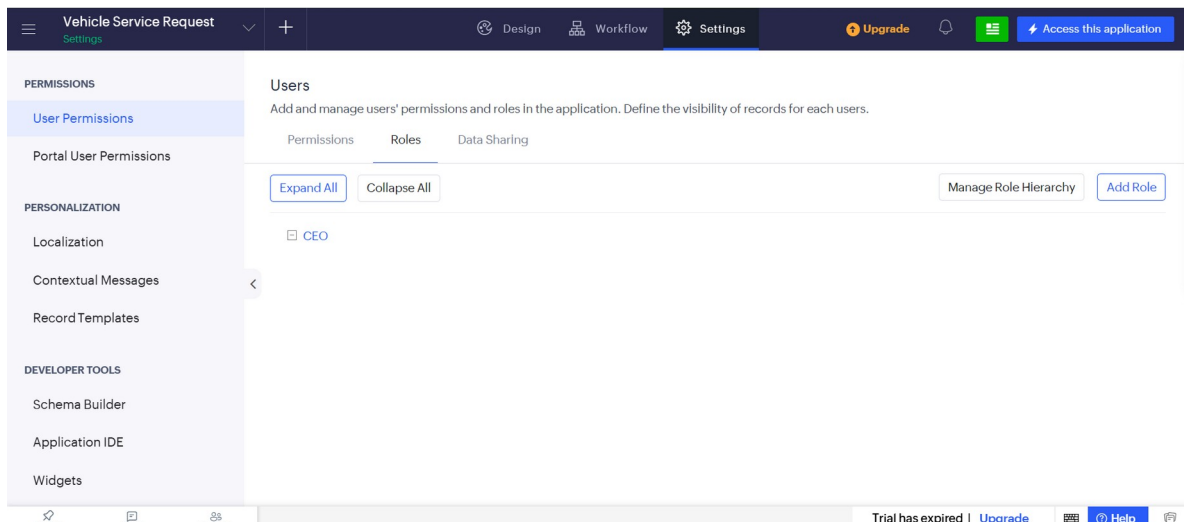


Figure 8.4: Roles and Role Hierarchy

8. **Attempt to configure an additional role:** Select Add Role and enter the details of the role to be created: Name – Service Manager; Reports to – CEO; Description – Manages vehicle service requests. The Share Data with Peers option remains unselected.

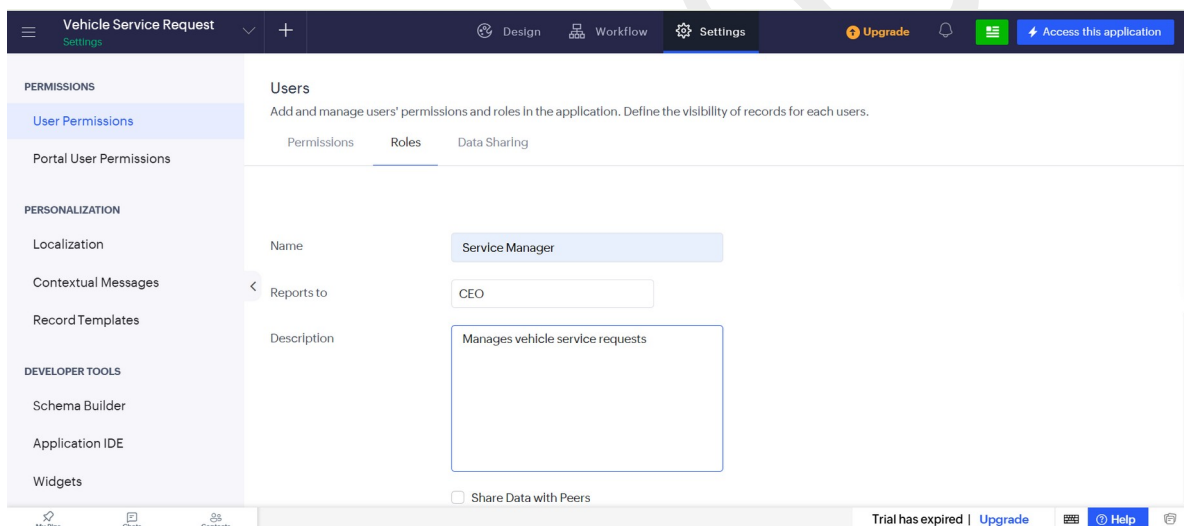


Figure 8.5: Service Manager Role Configuration (Attempted – Not Saved)

9. **Upgrade restriction:** On attempting to save the Service Manager role, Zoho Creator displayed an upgrade request. The application shows the status “Trial has expired” with an Upgrade option, and the current plan does not permit the creation of an additional role. Consequently, the Service Manager role was not saved, and CEO remains the only role in the application.
10. **Configure the Data Sharing Rule:** Select the Data Sharing tab and choose New Share Rule. In the New Share Rule panel, provide the rule name, select the form Vehicle Service Request, set Records shared from to the CEO role, set Records shared to to the user INDHUMATHY K, and set Level of access to Read/Write.
11. **Save the Data Sharing Rule:** Select Save. The rule is added to the Data Sharing list, as shown below.

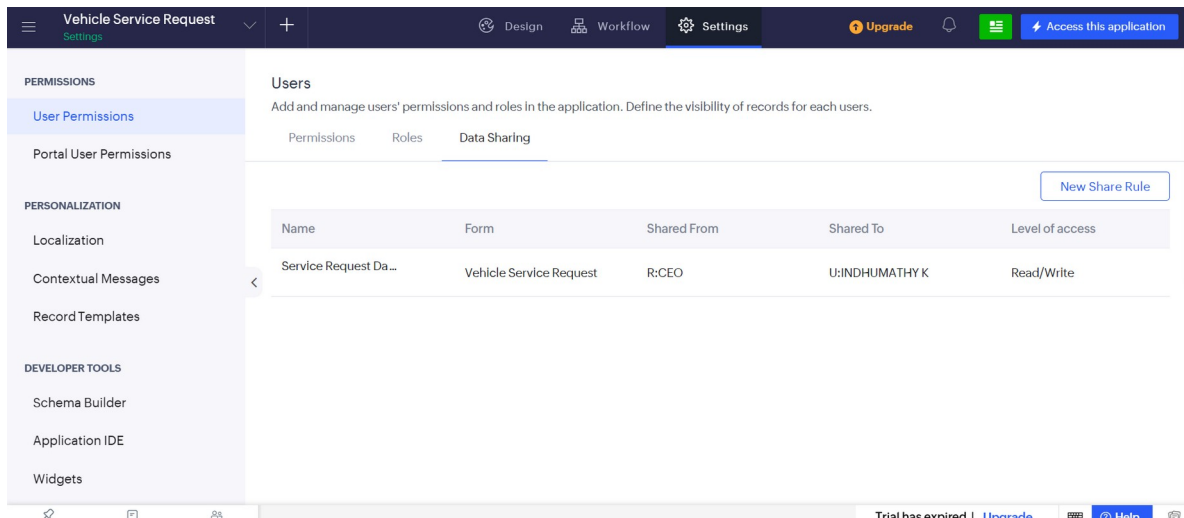


Figure 8.6: Configured Data Sharing Rule

RBAC Working

In a role-based design, users are placed into roles, and the roles are arranged in a hierarchy according to the organisational structure. The intended structure for the Vehicle Service Request application is a two-level hierarchy:

Level	Role	Intended Function
1 (Top)	CEO	Overall authority; able to view the records of roles reporting to it.
2	Service Manager	Manages vehicle service requests; reports to the CEO.

Roles actually available in the application: Only the default CEO role exists in the application. The Service Manager role was configured only up to the point of an attempted save and was not created, because the current plan does not allow an additional role. Accordingly, the Service Manager level in the table above represents the intended structure only, and no users were assigned to different roles.

Conceptually, once the Service Manager role is available, the hierarchy would operate as follows:

- Users assigned to the Service Manager role would work with service request records according to their permission set.
- Because the Service Manager role reports to the CEO, the CEO role, being higher in the hierarchy, would be able to view the records belonging to the Service Manager role.
- Users in the lower role would not automatically gain access to the records of the higher role, unless a data-sharing rule is defined for that purpose.
- Permission sets (Read or Write) would further limit the operations that each user may carry out on the application components.

Data Sharing

Role hierarchy alone provides only vertical visibility of records. Data sharing rules are used to grant additional access across roles or users. In this experiment, a share rule was configured for the Vehicle Service Request form with the following effect:

CEO → INDHUMATHY K → Read/Write

- The records of the Vehicle Service Request form belonging to the CEO role are the source of the sharing (Shared From: R:CEO).
- The user INDHUMATHY K is the recipient of the shared records (Shared To: U:INDHUMATHY K).
- The level of access is Read/Write, so the recipient can both view and modify the shared records.

The saved rule appears in the Data Sharing list (Figure 8.6) with the Form, Shared From, Shared To and Level of access columns. This demonstrates that record visibility can be extended beyond the default hierarchy in a controlled manner.

Result

The following were successfully carried out in the Vehicle Service Request application:

- The User Permissions section was accessed through Settings, and the available permission sets (Read and Write) were studied.
- The default CEO role and the Role Hierarchy interface were explored.
- A Data Sharing Rule was configured and saved for the Vehicle Service Request form, sharing records from the CEO role to the user INDHUMATHY K with Read/Write access.

The creation of an additional custom role (Service Manager, reporting to the CEO) was attempted; however, saving it was restricted by the current Zoho Creator plan, which requested an upgrade. Hence, only the default CEO role exists in the application.

Conclusion

The experiment demonstrated the user-management features of Zoho Creator, namely permissions, roles, role hierarchy and data sharing, and how they together form a role-based access control mechanism. The Read and Write permission sets, the default CEO role and the Role Hierarchy interface were examined, and a Data Sharing Rule granting Read/Write access to the Vehicle Service Request records was configured successfully. Although the creation of the Service Manager role was limited by the plan restrictions of the account, the concept of hierarchical role design was understood. The experiment shows that access to application data can be controlled in a structured and secure way through RBAC.